

Analytic Geometry-1st Quiz

1) Find the equation of the circle **tangent to** the line $d = 2x + y - 2 = 0$ and passing through the points $P(1,0)$ and $K(3,-2)$.

2) Let $(6, -4)$ and $B(9,2)$ be two points in the orthogonal coordinate system XOY with central $O(0,0)$, and $X''AY''$ be the oblique system where;

- **A is the central point,**
- **The X'' axis is on vector $\overrightarrow{AB}$,**
- **The Y'' axis is parallel to the Y axis,**

a) Obtain the transition formulas from orthogonal coordinate system XOY to oblique coordinate system $X''AY''$.

b) Let $M(7,1)$ be a point in orthogonal coordinate system XOY . Find the **new coordinates** of M in the oblique coordinate system $X''AY''$.

3) The planes $D1 : x + 2y - z = 1$ and the line $d: \begin{matrix} x = 1 + t, \\ y = 2 - t, \\ z = 1 - t \end{matrix}$ are given.

a) Find the equation of the plane which is perpendicular to the **intersection line** of $D1$ and $D2$ planes and also passing through the point $P(1, -2, 1)$.

b) Check whether the plane **$D1$ is parallel to line d or not.**

4) Find the **parametric equation** of the intersection line of the planes

$D1: x + y = 1$ and

$D2: 2x + y - 2z = 2$.

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Good Luck.